

Laboratory Work 2:Ersinbek Erdana

Normalization

1. Normal Form Violation

The original table is:

Booking_receipt: booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)

The table violates **1NF** because seat_numbers can contain multiple values in one cell, for example: 12A, 12B. In 1NF, each attribute must contain only one atomic value.

It also contains dependencies that must be separated to achieve **2NF and 3NF**, such as:

booking_id → flight_number, ticket_price

passport_number → passenger_full_name

flight_number → airline_name

airport_id → airport_name, city

2. Normalization Step by Step

1NF

Separate seat_numbers into individual values.

Booking_receipt_1NF

(booking_id, passport_number, seat_number), passenger_full_name, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price

Primary Key: (**booking_id, passport_number, seat_number**)

2NF

Remove partial dependencies by separating passengers and bookings.

Passenger

passport_number, full_name

Booking

booking_id, flight_number, ticket_price

Booking_passenger

(booking_id, passport_number), seat_number

Foreign Keys:

booking_id → Booking

passport_number → Passenger

3NF

Remove transitive dependencies by separating flights, airlines, and airports.

Airline

airline_id, airline_name

Airport

airport_id, airport_name, city

Flight

flight_number, airline_id, departure_airport_id, arrival_airport_id

Foreign Keys:

airline_id → Airline

departure_airport_id → Airport

arrival_airport_id → Airport

The final database is in **3NF** because non-key attributes depend only on the primary key.

3. Functional Dependencies

The main functional dependencies are:

passport_number → full_name

booking_id → flight_number, ticket_price

flight_number → airline_id, departure_airport_id, arrival_airport_id

airline_id → airline_name

airport_id → airport_name, city

These dependencies were used to separate the tables during normalization.

4. Flights Table and 3NF

The primary key of the Flights table is **flight_id**.

Functional dependencies include:

flight_id → flight_number

flight_id → departing_airport_id

flight_id → arriving_airport_id

flight_id → airline_id

flight_id → departure_time

flight_id → arrival_time

There is no transitive dependency if airline and airport information is stored in separate tables.

For example:

flight_id → airline_id → airline_name

The airline_name is stored in the Airline table, not in Flights.

Therefore, the **Flights table is in 3NF**.

DDL:

The screenshot shows a PostgreSQL IDE interface. On the left, a tree view displays the database structure, including 'Databases (1)' with 'postgres', and 'Schemas (1)' with 'public'. The 'public' schema is expanded, showing various database objects like 'Aggregates', 'Collations', 'Domains', 'FTS Configurations', 'FTS Dictionaries', 'FTS Parsers', 'FTS Templates', 'Foreign Tables', 'Functions', 'Materialized Views', 'Operators', 'Procedures', 'Sequences', and 'Tables (10)'. The main query editor displays the following SQL code:

```
1
2 -- 5. Rename airline_info to airline
3
4 ALTER TABLE airline_info
5 RENAME TO airline;
6 SELECT * FROM airline;
```

Below the query editor, the 'Data Output' tab is active, showing the structure of the 'airline' table. The table has the following columns:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at	info
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone	character varying (50)

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Query

Query History

Scratch Pad

```
1
2  -- 6. Rename price to ticket_price
3
4  ALTER TABLE booking
5  RENAME COLUMN price TO ticket_price;
6  SELECT * FROM booking;
```

Data Output Messages Notifications

booking_id	flight_id	passenger_id	booking_platform	created_at	updated_at	status	ticket_price
[PK] integer	integer	integer	character varying (50)	timestamp without time zone	timestamp without time zone	character varying (50)	numeric (7,2)

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Query

Query History

Scratch Pad

```
1
2  -- 7. Change departing_gate type to TEXT
3
4  ALTER TABLE flights
5  ALTER COLUMN departing_gate TYPE TEXT;
6
7  SELECT * FROM flights;
```

Data Output Messages Notifications

flight_id	sch_departure_time	sch_arrival_time	departing_airport_id	arriving_airport_id	departing_gate	arriving_gate	airline_id	act_departure
[PK] integer	timestamp without time zone	timestamp without time zone	integer	integer	text	character varying (50)	integer	timestamp wit

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Query

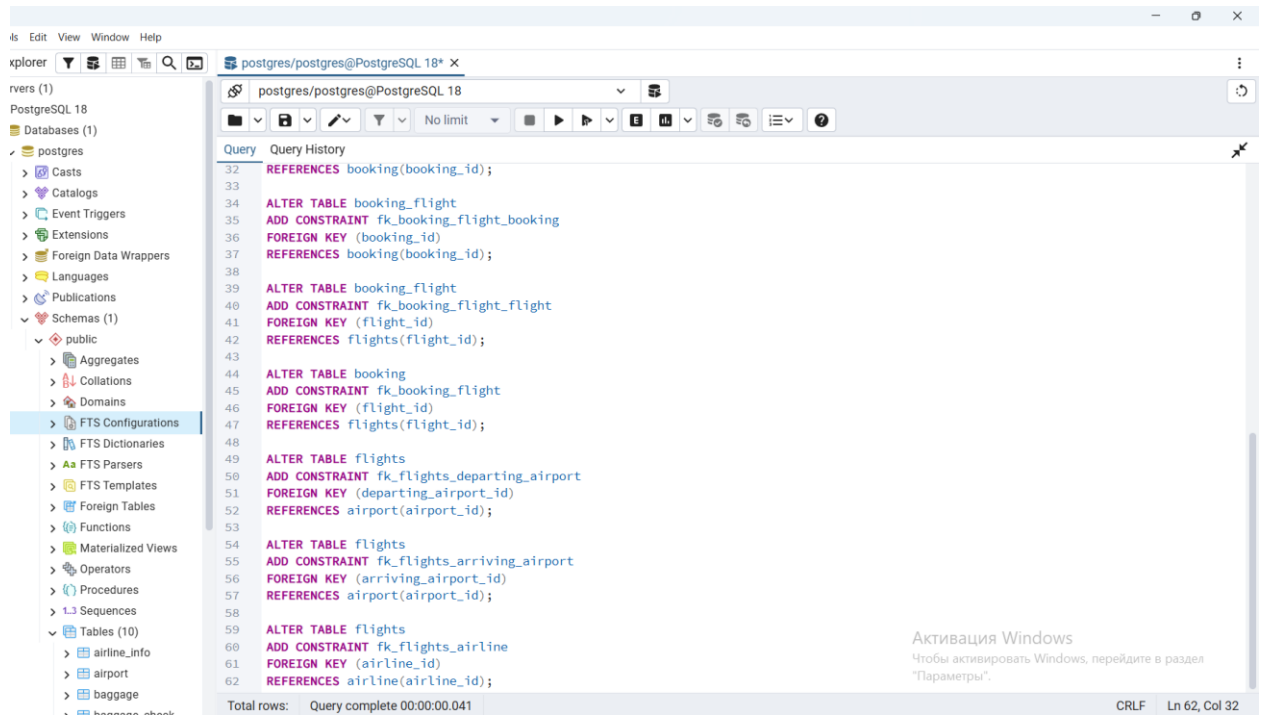
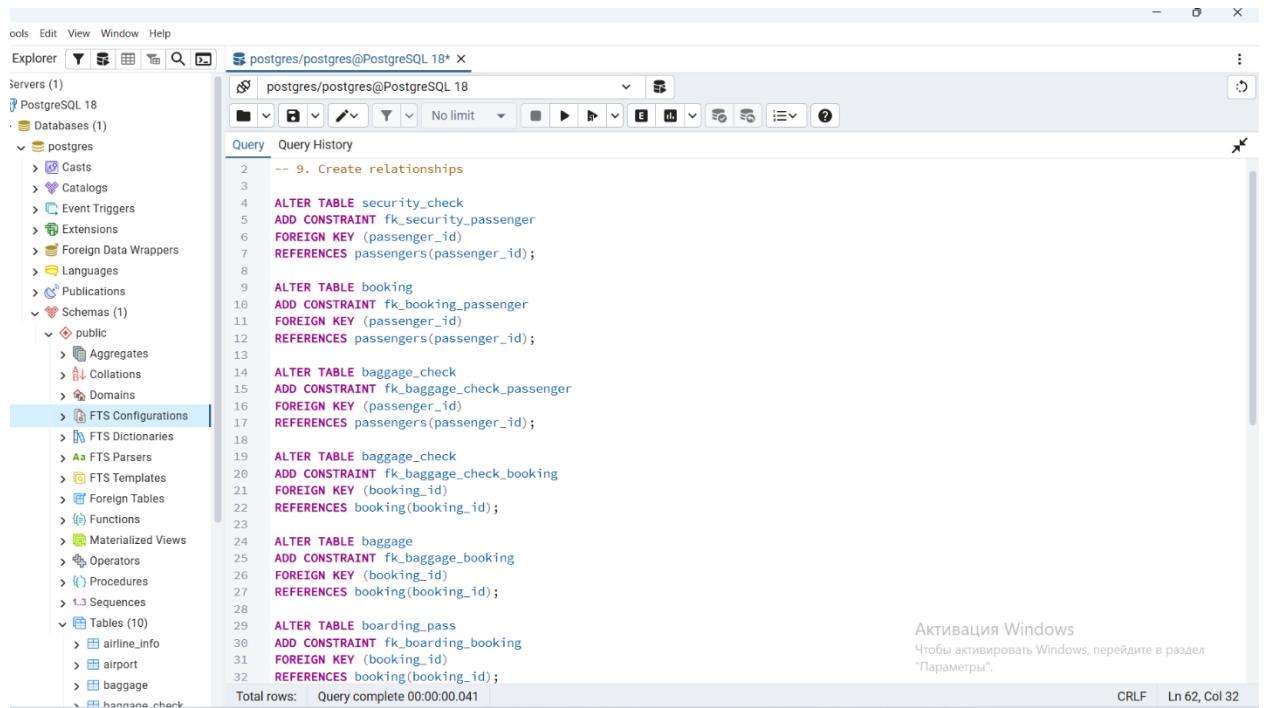
Query History

Scratch Pad

```
1
2  -- 8. Drop info column
3
4  ALTER TABLE airline
5  DROP COLUMN info;
6  SELECT * FROM airline;
```

Data Output Messages Notifications

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone



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Query Query History

```
2 CREATE TABLE airline_info (  
3     airline_id INT NOT NULL,  
4     airline_code VARCHAR(30) NOT NULL,  
5     airline_name VARCHAR(50) NOT NULL,  
6     airline_country VARCHAR(50) NOT NULL,  
7     created_at TIMESTAMP NOT NULL,  
8     updated_at TIMESTAMP NOT NULL,  
9     info VARCHAR(50) NOT NULL,  
10    PRIMARY KEY (airline_id)  
11 );  
12  
13 CREATE TABLE airport (  
14     airport_id INT NOT NULL,  
15     airport_name VARCHAR(50) NOT NULL,  
16     country VARCHAR(50) NOT NULL,  
17     state VARCHAR(50) NOT NULL,  
18     city VARCHAR(50) NOT NULL,  
19     created_at TIMESTAMP NOT NULL,  
20     updated_at TIMESTAMP NOT NULL,  
21     PRIMARY KEY (airport_id)  
22 );  
23  
24 CREATE TABLE baggage_check (  
25     baggage_check_id INT NOT NULL,  
26     check_result VARCHAR(50) NOT NULL,  
27     created_at TIMESTAMP NOT NULL,  
28     updated_at TIMESTAMP NOT NULL,  
29     booking_id INT NOT NULL,  
30     passenger_id INT NOT NULL,  
31     PRIMARY KEY (baggage_check_id)  
32 );
```

Total rows: Query complete 00:00:00.041

Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".

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Query Query History

```
80 departing_gate VARCHAR(50) NOT NULL,  
81 arriving_gate VARCHAR(50) NOT NULL,  
82 airline_id INT NOT NULL,  
83 act_departure_time TIMESTAMP NOT NULL,  
84 act_arrival_time TIMESTAMP NOT NULL,  
85 created_at TIMESTAMP NOT NULL,  
86 updated_at TIMESTAMP NOT NULL,  
87 PRIMARY KEY (flight_id)  
88 );  
89  
90 CREATE TABLE passengers (  
91     passenger_id INT NOT NULL,  
92     first_name VARCHAR(50) NOT NULL,  
93     last_name VARCHAR(50) NOT NULL,  
94     date_of_birth DATE NOT NULL,  
95     gender VARCHAR(50) NOT NULL,  
96     country_of_citizenship VARCHAR(50) NOT NULL,  
97     country_of_residence VARCHAR(50) NOT NULL,  
98     passport_number VARCHAR(20) NOT NULL,  
99     created_at TIMESTAMP NOT NULL,  
100    updated_at TIMESTAMP NOT NULL,  
101    PRIMARY KEY (passenger_id)  
102 );  
103  
104 CREATE TABLE security_check (  
105     security_check_id INT NOT NULL,  
106     check_result VARCHAR(20) NOT NULL,  
107     created_at TIMESTAMP NOT NULL,  
108     updated_at TIMESTAMP NOT NULL,  
109     passenger_id INT NOT NULL,  
110     PRIMARY KEY (security_check_id)
```

Total rows: Query complete 00:00:00.041

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Чтобы активировать Windows, перейдите в раздел "Параметры".